Tutorial Sheet 1 Solutions

Q. No.	max.	min.	sup.	inf.	bounded above	bounded below	bounded
a) S_1	1	does not exist	$1 \in S_1$	$0 \notin S_1$	✓	✓	✓
b) S_2	does not exist	-1	$0 \notin S_2$	$-1 \in S_2$	✓	✓	✓
c) S_3	$1/2$	-1	$1/2 \in S_3$	$-1 \in S_3$	✓	✓	✓
d) S_4	$3/2$	0	$\frac{3}{2} \in S_4$	$0 \in S_4$	✓	✓	✓
e) S_5	does not exist	does not exist	$1 \notin S_5$	$-1 \notin S_5$	✓	✓	✓
f) S_6	does not exist	does not exist	$2 \notin S_6$	$1 \notin S_6$	✓	✓	✓
g) S_7	does not exist	does not exist	does not exist	does not exist	×	×	×
h) S_8	does not exist	does not exist	$\sqrt{2} \notin S_8$	$-\sqrt{2} \notin S_8$	✓	✓	✓
i) S_9	2	does not exist	$2 \in S_9$	$0 \notin S_9$	✓	✓	✓
j) S_{10}	2	does not exist	$2 \in S_{10}$	$-1 \notin S_{10}$	✓	✓	✓

$S_1 = \{1, \frac{1}{2}, \frac{1}{3}, \cdots\}$, $S_2 = \{-1, -\frac{1}{2}, -\frac{1}{3}, \cdots\}$

$S_3 = \{-1, -\frac{1}{3}, -\frac{1}{5}, \cdots\} \cup \{\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \cdots\}$

$S_4 = \{0, 1-\frac{1}{3}, 1-\frac{1}{5}, \cdots\} \cup \{1+\frac{1}{2}, 1+\frac{1}{4}, 1+\frac{1}{6}, \cdots\}$

$S_5 = (-1, 1)$, $S_6 = (1, 2)$, $S_7 = (-\infty, 1) \cup (2, \infty)$

$S_8 = \{x \in \mathbb{Q} \mid -\sqrt{2} \leq x \leq \sqrt{2}\}$ $S_9 = \{0 < \frac{1}{m} + \frac{1}{n} \leq 2 : m, n \in \mathbb{N}\}$

$S_{10} = \{1 + (-1)^m : m, n \in \mathbb{N}\}$

$a_{10} = \{-1 < (-1) + \frac{1}{n} \leq 2 \cdot n, n \in \mathbb{N}\}$

2) a) $\{1,2,3,4\}$

b) $\mathbb{Z}$, $\mathbb{R}$

c) $\mathbb{N}$

d) $(-\infty, 0]$